

SHIPRA SINGH

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Profile

I am a Biotechnology postgraduate with a strong foundation in molecular biology and microbiology. I have hands-on experience in techniques such as PCR, gel electrophoresis, and primer design for SNPs and rslDs, as well as in handling media and autoclaving. I completed my dissertation in the Genetics and Genomics Department at RIMS Hospital, Ranchi, titled "Optimization of the DNA Purification Protocol Utilizing Magnetic Beads." The project focused on DNA purification using magnetic beads, where I gained expertise in DNA/RNA isolation both manually and through kits, as well as purification using magnetic beads.

Work Experience

Dissertation Research

Genetics and Genomics Department, RIMS Hospital, Ranchi
March 2024 – September 2024

• Conducted Dissertation on "**Optimization of the Purification Protocol Utilizing Magnetic Beads.**" under the guidance of Dr. Sunil Shakya.

Gained expertise in DNA purification using magnetic beads, improving sample quality for downstream applications.

• Designed primers for SNPs and rslDs to study genetic variations.

• Applied manual protocols for DNA isolation from blood samples and optimized workflows for molecular diagnostics.

• Performed PCR and gel electrophoresis for DNA integrity analysis.

• Conducted DNA and RNA isolation using commercial kits, ensuring high-quality nucleic acid extraction for experimental purposes. I have one year of experience as a home tutor, teaching students from 4th to 8th standard.

Education

- **Master of Science in Biotechnology**

- Ranchi University, 2024

- CGPA: 8.44

- **Bachelor of Science in zoology**

- Vinoba Bhave University, 2022

- CGPA: 7.46

Technical Skills

- **Molecular Techniques:** PCR, Gel Electrophoresis, Primer Designing, SNP Primer Development
- **DNA/RNA Handling:** DNA Isolation (manual and kit-based), DNA Purification, RNA Isolation (kit-based)
- **Microbiology Lab-** Bacterial and Plant Tissue culture.
- **Creative Designing:** Canva, Adobe Photoshop
- **Software and Tools:** AI Tools: ChatGPT, Perplexity, ChatPDF, Research Rabbit
- **Bioinformatics Applications:** BLAST, FASTA,
- **Prepared and sterilized nutrient media**
- **Serial Dilution**

Activities

- **Attended a UNDP webinar on Youth Co:Lab National Innovation Challenge 2024-2025**, in collaboration with AssisTech Foundation (**ATF**), focused on enhancing access to opportunities and well-being of persons with disabilities, 2024.
- **Online Bioinformatics Internship, Biotechnika, 2023**
- **Participated in webinars on CRISPR-Cas9 and AI in Drug Discovery (Biotechnika)**
- **Attended a Sustainable Agricultural Program on Kisan Divas, 24th December 2023**
- **Presented a poster titled "Brain-Gut Axis" during World Microorganism Day 2023.**
- **Participated in a research-oriented writing competition as part of the World Microorganism Day 2023 event**